HW2

Tuesday, February 10, 2026 9:12 PM

1) ISL 4.8.1 ~ Prove Logistic Func Rep = Logit rep. for logistic regression model

$$p(X) = \frac{e^{\beta_0+\beta_1 X}}{1+e^{\beta_0+\beta_1 X}} \quad ; \text{ let } z = \beta_0+\beta_1 X \rightarrow p = \frac{e^z}{1+e^z}$$

p is prob event occurs

* logit is defined by odds = $P/1-p$; we know form of p & want to manip. form of $1-p$

$$\rightarrow 1-p = 1-\frac{e^z}{1+e^z} = \frac{1+e^z}{1+e^z} - \frac{e^z}{1+e^z} = \frac{1}{1+e^z}$$

$$\therefore \frac{p}{1-p} = \left(e^z/_{1+e^z}\right) / \left(1/_{1+e^z}\right) = \left(e^z/_{1+e^z}\right) \cdot \left(1+e^z/_1\right) = e^z$$

$$\therefore \frac{p(X)}{1-p(X)} = e^{\beta_0+\beta_1 X}$$; then taking log of both sides to derive logit form

$$\rightarrow \boxed{\log\left(p(X)/_{1-p(X)}\right)} = \log\left(e^{\beta_0+\beta_1 X}\right) = \boxed{\beta_0+\beta_1 X}$$; since $\log(e^a) = a$

2) ISL 4.8.6

For this problem the model is construed as :

$$Y = \log\left(\frac{p(X)}{1-p(X)}\right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 \xrightarrow{\text{plug in est}} -6 + 0.05X_1 + 1X_2$$

$\therefore$ the logistic function representation takes the form

$$p(X) = \frac{e^{-6+0.05X_1+X_2}}{1+e^{-6+0.05X_1+X_2}}$$

a) Therefore to obtain odds for a student w/ 40hr study + GPA = 3.5 to obtain A we plug in 40 for X_1 & 3.5 for X_2

$$\rightarrow p(X) = \frac{e^{-6+0.05(40)+3.5}}{1+e^{-6+0.05(40)+3.5}} = \frac{e^{-0.5}}{1+e^{-0.5}} \approx 0.3775$$

$\therefore$ The prob. a student who studies for 40hr & has a GPA of 3.5 gets an A in the class is approx. 37.75%

b) the prob. can be rep. as :

$$\frac{e^{-6+0.05X+3.5}}{1+e^{-6+0.05X+3.5}} = 0.5 \quad \& \quad \log\left(0.5/_{1-0.5}\right) = -6+0.05X+3.5 = \varnothing$$

& solving for X : $2.5 = 0.05X \rightarrow X = 50$

$\therefore$ The student in part a needs to study 50 hrs. to have a 50% chance of getting an A in the class.

3) ISL 4.8.9

a) odds : $P/1-p = 0.37$ from problem setup

$$\rightarrow 0.37(1-p) = p \rightarrow 0.37 = 1.37p \rightarrow p = \frac{0.37}{1.37} \approx 0.27$$

Given odds of 0.37 for defaulting on credit card payment, approx. 27% on average will default

b) Given 16% prob/chance of default :

$$\text{odds} = P/1-p = 0.16/(1-0.16) = 4/21 \approx 0.19$$

The odds she will default is approx. 19%